Q1.	Develop an interactive Matplotlib program for the parabola $y=ax^2$ with the following features: 1. Create a Slider for (a), ranging from (-2) to (2). 2. Set (a=1) initially. 3. Update the parabola whenever the slider is moved. 4. Add a **Reset** button. 5. Use set_ydata() to update the existing curve. 6. Label the axes and add a suitable title.
Q2	**RadioButtons for Different Mathematical Functions** **Question:** Develop an interactive Matplotlib program using RadioButtons that allows the user to select one of the following: • Linear: (y=x) • Quadratic: ($y=x^2$) • Cubic: ($y=x^3$) • Sine: (y=sin x) When the user selects a function, the corresponding curve should be displayed on the same Axes.
Q3	**CheckButtons for Curve Visibility** **Question:** Write a Python program using CheckButtons to plot the following three functions: y=sin x ,y=cos x , y=sin(2x) Provide independent checkboxes for each function so that the user can show or hide any combination of curves.
Q4.	Consider the parametric representation $x=t, y=at^2$ Write a Python program using a Slider to change (a) interactively between (-2) and (2). The program should: • generate (t) using np.linspace(), • calculate (x) and (y), • plot the parametric curve, • update the curve when (a) changes, • use fig.canvas.draw_idle().
Q5	Develop an interactive Matplotlib program that provides: 1. A RadioButtons widget to select between $y=x^2$ and $y=x^3$. 2. A Slider to change the coefficient (a). 3. A Button to reset the coefficient to its initial value. 4. Appropriate axis labels, title, and grid.